

X100/302

NATIONAL
QUALIFICATIONS
2009

THURSDAY, 21 MAY
10.50 AM – 12.00 NOON

MATHEMATICS
HIGHER
Paper 2

Read Carefully

- 1 **Calculators may be used in this paper.**
- 2 Full credit will be given only where the solution contains appropriate working.
- 3 Answers obtained by readings from scale drawings will not receive any credit.



FORMULAE LIST

Circle:

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r .

Scalar Product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

or $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

Trigonometric formulae: $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2\cos^2 A - 1$$

$$= 1 - 2\sin^2 A$$

Table of standard derivatives:

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals:

$f(x)$	$\int f(x) dx$
$\sin ax$	$-\frac{1}{a} \cos ax + C$
$\cos ax$	$\frac{1}{a} \sin ax + C$

ALL questions should be attempted.

Marks

1. Find the coordinates of the turning points of the curve with equation $y = x^3 - 3x^2 - 9x + 12$ and determine their nature.

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2. Functions f and g are given by $f(x) = 3x + 1$ and $g(x) = x^2 - 2$.

(a) (i) Find $p(x)$ where $p(x) = f(g(x))$.

(ii) Find $q(x)$ where $q(x) = g(f(x))$.

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(b) Solve $p'(x) = q'(x)$.

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3. (a) (i) Show that $x = 1$ is a root of $x^3 + 8x^2 + 11x - 20 = 0$.

(ii) Hence factorise $x^3 + 8x^2 + 11x - 20$ fully.

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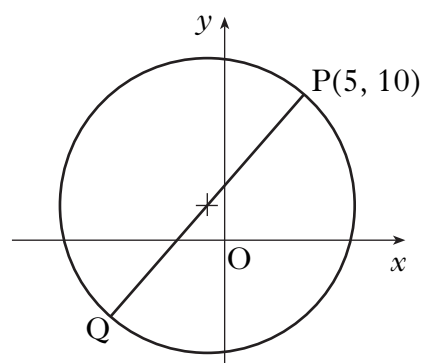
(b) Solve $\log_2(x + 3) + \log_2(x^2 + 5x - 4) = 3$.

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4. (a) Show that the point $P(5, 10)$ lies on circle C_1 with equation $(x + 1)^2 + (y - 2)^2 = 100$.

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(b) PQ is a diameter of this circle as shown in the diagram. Find the equation of the tangent at Q .



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(c) Two circles, C_2 and C_3 , touch circle C_1 at Q .

The radius of each of these circles is twice the radius of circle C_1 .

Find the equations of circles C_2 and C_3 .

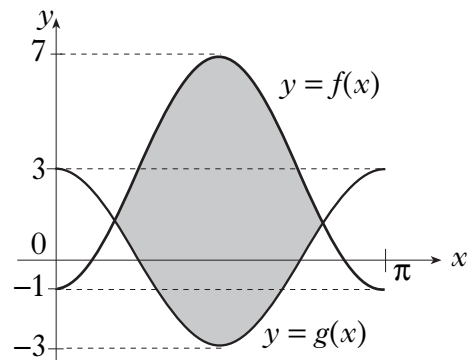
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[Turn over

5. The graphs of $y = f(x)$ and $y = g(x)$ are shown in the diagram.

$f(x) = -4 \cos(2x) + 3$ and $g(x)$ is of the form $g(x) = m \cos(nx)$.

- (a) Write down the values of m and n .
 (b) Find, correct to one decimal place, the coordinates of the points of intersection of the two graphs in the interval $0 \leq x \leq \pi$.
 (c) Calculate the shaded area.



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6. The size of the human population, N , can be modelled using the equation $N = N_0 e^{rt}$ where N_0 is the population in 2006, t is the time in years since 2006, and r is the annual rate of increase in the population.

- (a) In 2006 the population of the United Kingdom was approximately 61 million, with an annual rate of increase of 1.6%. Assuming this growth rate remains constant, what would be the population in 2020?
 (b) In 2006 the population of Scotland was approximately 5.1 million, with an annual rate of increase of 0.43%.

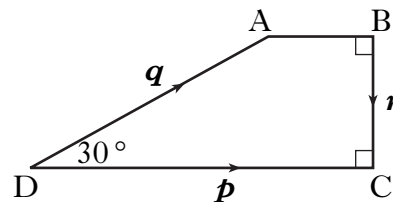
Assuming this growth rate remains constant, how long would it take for Scotland's population to double in size?

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7. Vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ are represented on the diagram shown where angle $ADC = 30^\circ$.

It is also given that $|\mathbf{p}| = 4$ and $|\mathbf{q}| = 3$.

- (a) Evaluate $\mathbf{p} \cdot (\mathbf{q} + \mathbf{r})$ and $\mathbf{r} \cdot (\mathbf{p} - \mathbf{q})$.
 (b) Find $|\mathbf{q} + \mathbf{r}|$ and $|\mathbf{p} - \mathbf{q}|$.



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[END OF QUESTION PAPER]